i.e., $\int_0^{a_1} f(x)dx = \frac{1}{2}$

i.e., $\int_0^{a_1} 5x^4 dx = \frac{1}{2}$

$$5\left[\frac{x^5}{5}\right]_0^{a_1} = \frac{1}{2}$$

$$a_1 = (0.5)^{\frac{1}{5}}$$

(ii) $P[X > a_2] = 0.05$

$$\int_{a_2}^{1} f(x)dx = 0.05$$

$$\int_{a_2}^{1} 5x^4 dx = 0.05$$

$$5\left[\frac{x^5}{5}\right]_{a_2}^{1} = 0.05$$

$$a_2 = [0.95]^{\frac{1}{5}}$$

Continuous distribution function

Definition 6.7

If X is a continuous random variable with the probability density function $f_X(x)$, then the function $F_X(x)$ is defined by

$$F_X(x) = P[X \le x] = \int_{-\infty}^{x} f(t)dt, -\infty < x < \infty$$

is called the distribution function (d.f) or sometimes the cumulative distribution function (c.d.f) of the continuous random variable X.

Properties of cumulative distribution function

The function $F_X(x)$ or simply $F(x)$ has the following properties

(i) $0 \le F(x) \le 1, -\infty < x < \infty$

(ii) $F(-\infty) = \lim_{x \to -\infty} F(x) = 0$ and $F(+\infty) = \lim_{x \to \infty} F(x) = 1$.

(iii) $F(\cdot)$ is a monotone, non-decreasing function; that is, $F(a) \le F(b)$ for $a < b$.

(iv) $F(\cdot)$ is continuous from the right; that is, $\lim_{h \to 0} F(x+h) = F(x)$.

(v) $F'(x) = \frac{d}{dx}F(x) = f(x) \ge 0$

(vi) $F'(x) = \frac{d}{dx}F(x) = f(x) \Rightarrow dF(x) = f(x)dx$, $dF(x)$ is known as probability differential of X.

(vii) $P(a \le x \le b) = \int_a^b f(x)dx = \int_{-\infty}^{b} f(x)dx - \int_{-\infty}^{a} f(x)dx$

$= P(X \le b) - P(X \le a)$

$= F(b) - F(a)$

Example 6.10

Suppose, the life in hours of a radio tube has the following p.d.f

$$f(x) = \begin{cases} \frac{100}{x^2}, when\, x \ge 100 \\ 0, \quad when\, x < 100 \end{cases}$$

Find the distribution function.

Solution:

$$F(x) = \int_{-\infty}^{x} f(t)dt$$

$$= \int_{100}^{x} \frac{100}{t^2}dt, x \ge 100$$

$$= \left[\frac{100}{-t}\right]_{100}^{x}, x \ge 100$$

$$F(x) = \left[1 - \frac{100}{x}\right], x \ge 100$$

Example 6.11

The amount of bread (in hundreds of pounds) x that a certain bakery is able to sell in a day is found to be a numerical valued random phenomenon, with a probability function specified by the probability density function $f(x)$ is given by

$$f(x) = \begin{cases} Ax, for\, 0 \le x < 10 \\ A(20-x), for\, 10 \le x < 20 \\ 0, \qquad otherwise \end{cases}$$

(a) Find the value of A.

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